Sanketh Edara

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Education

Purdue University | Bachelor of Science West Lafayette, IN
Dual Major: *Computer Science and Data Science* ***Minors:*** *Finance, Mathematics* *May 2027*

Relevant Coursework: Data Structures and Algorithms, Analysis of Algorithms, Data Mining and Machine Learning, Relational Databases, Systems Programming, Computer Architecture, Artificial Intelligence, Programming in C, OOPS

Experience

Interim Engineering Intern | C++, Embedded Systems, Digital Signal Processing May 2025 – Aug 2025
Qualcomm
- Built a real-time **C++** sensor fusion pipeline for XR orientation tracking using Extended Kalman Filter logic on IMU streams.
- Integrated production code with Qualcomm's sensor stack for reliable ingestion, processing, and debugging of device data.
- Optimized runtime with Hexagon Vector Extensions, reducing computational latency and improving performance by **30%**.
- Cut average power usage by up to **90%** by offloading compute-intensive processing from CPU to DSP.
- Partnered across platform teams to profile bottlenecks, validate outputs, and ship efficient embedded software.

Full Stack Engineering Co-Op | Python, Typescript, Flask, Firebase, NoSQL, APIs Aug 2023 – May 2024
Caterpillar
- Built internal **Python** forecasting services for EV charging operations and exposed outputs through APIs and data pipelines.
- Developed JSON-based transformation pipelines that improved integration between backend systems and dashboards.
- Prototyped an AI analytics assistant over EV telemetry data to surface trends and reduce manual analysis time.
- Evaluated NBEATS, ARIMA, and LSTM models with scenario-based validation to improve production reliability.
- Delivered software that improved fleet planning efficiency and reduced operational uncertainty in EV workflows.

Software Engineering Intern | Python, NLP, TF-IDF/BERT May 2024 – Aug 2024
QXO
- Built an AI-powered speech evaluation platform using **Python**, Whisper, and NLP pipelines for scoring and feedback.
- Designed retrieval and embedding workflows with TF-IDF and BERT to improve output consistency and scoring accuracy.
- Developed a modular backend pipeline for transcription, feature extraction, evaluation, and feedback delivery.
- Automated evaluation workflows with AI tooling, reducing manual review effort by **75%**.
- Contributed to product architecture combining model inference, backend orchestration, and user-facing output generation.

Projects

Nebulus (Multi-agent AI COO) | Next.js, TypeScript, FastAPI, Python, Supabase, PostgreSQL, Redis, Celery
- Built a full stack multi-agent AI COO with a **Next.js** control layer, **FastAPI** backend, and approval-aware operator workflows.
- Designed shared context, execution, and task routing across specialized agents with backend and cloud-based workflow state.
- Used **PostgreSQL**, **Redis**, and **Celery** for shared state, job queues, retries, and background orchestration.
- Integrated **Supabase** for real-time sync, object storage, and **SQL**-backed workflow visibility across teams.

Evalon (Chess Evaluation Tool) | Next.js, TypeScript, Chess.js, Stockfish
- Built a full stack chess copilot that converts PGN games into structured review, engine analysis, and training workflows.
- Developed **Next.js** interfaces for PGN upload, position review, move-by-move breakdowns, and critical mistake exploration.
- Used **Chess.js** for move validation, board-state management, and game parsing across frontend analysis and review flows.
- Integrated **Stockfish** to generate engine evaluations, opening insights, and critical turning points for systematic game review.

SignSense (American Sign Language Translator) | React, Python, MediaPipe, Auth0, Tailwind CSS
- Built a full stack ASL recognition platform for gesture understanding, guided practice, and interactive sign-based learning flows.
- Developed a **React** frontend for live recognition, learning modules, webcam interaction, and responsive user practice workflows.
- Used **MediaPipe** and **Python** to power hand tracking, landmark extraction, and backend recognition workflows for ASL input.
- Integrated **Auth0** for secure login, session management, and personalized access across recognition and learning experiences.

Technical Skills

Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, R
Frameworks & Tools: React, Node.js, FastAPI, Flask, Django, Git, GitHub, Docker, Linux/Unix, Jupyter, VS Code
Databases & Cloud: PostgreSQL, Firebase, Supabase, SQL, NoSQL, AWS (EC2, S3), GCP, Redis
Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Hugging Face, Matplotlib, Statsmodels
Concepts: Machine Learning, Artificial Intelligence, LLMs, Data Engineering, Backend Development, Fullstack Development, CI/CD Pipelines, Cloud Engineering, Agile Methodology, REST API
Systems & Performance: C/C++, Embedded Systems, Linux/Unix, Real-Time Systems
Robotics / Perception: Sensor Fusion, SLAM, Computer Vision, OpenCV, ROS